

Ekvivalentními úpravami dostaneme z pravých stran ze zadání levé strany ze zadání takto:

$$\begin{aligned}
\frac{1}{4}(|\mathbf{u} + \mathbf{v}|^2 - |\mathbf{u} - \mathbf{v}|^2) &= \frac{1}{4}(\langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle) = \\
&= \frac{1}{4}(\langle \mathbf{u} + \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u} + \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u} - \mathbf{v}, \mathbf{v} \rangle) = \\
&= \frac{1}{4}(\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle) = \\
&= \frac{1}{2}(\langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{2}(\overline{\langle \mathbf{u}, \mathbf{v} \rangle} + \langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{2}(2 \Re(\langle \mathbf{u}, \mathbf{v} \rangle)) = \Re(\langle \mathbf{u}, \mathbf{v} \rangle)
\end{aligned}$$

$$\begin{aligned}
\frac{1}{4}(|\mathbf{u} - i\mathbf{v}|^2 - |\mathbf{u} + i\mathbf{v}|^2) &= \frac{1}{4}(\langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle) = \\
&= \frac{1}{4}(\langle \mathbf{u} - i\mathbf{v}, \mathbf{u} \rangle - \langle \mathbf{u} - i\mathbf{v}, i\mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u} + i\mathbf{v}, i\mathbf{v} \rangle) = \\
&= \frac{1}{4}(\langle \mathbf{u}, \mathbf{u} \rangle - \langle i\mathbf{v}, \mathbf{u} \rangle - \langle \mathbf{u}, i\mathbf{v} \rangle + \langle i\mathbf{v}, i\mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle - \langle i\mathbf{v}, \mathbf{u} \rangle - \langle \mathbf{u}, i\mathbf{v} \rangle - \langle i\mathbf{v}, i\mathbf{v} \rangle) = \\
&= -\frac{1}{2}(\langle i\mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u}, i\mathbf{v} \rangle) = -\frac{1}{2}(-i\langle \mathbf{v}, \mathbf{u} \rangle + i\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{i}{2}(\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{u}, \mathbf{v} \rangle) = \\
&= \frac{i}{2}(-i \times 2 \Im(\langle \mathbf{u}, \mathbf{v} \rangle)) = \Im(\langle \mathbf{u}, \mathbf{v} \rangle)
\end{aligned}$$

V prvních krocích jsme využili znalost $|\mathbf{u} + \mathbf{v}|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle$. V druhých jsme dělali rozvoj dle SL2 z 8.14, v třetích dle (3) z 8.15. Ve čtvrtých jsme počítali, co se dalo. V pátém kroku druhé rovnosti jsme použili SL1 z 8.14 a (2) z 8.15. V pátém kroku první rovnosti a šestém kroku druhé rovnosti jsme díky znalosti SSS z 8.14 došli k součtu, respektive rozdílu čísla a jeho komplexně sdruženého čísla. Tam jsme použili:

$$z + \bar{z} = a + bi + a - bi = 2a = 2 \Re(z)$$

$$z - \bar{z} = a + bi - (a - bi) = 2bi = 2i \Im(z).$$